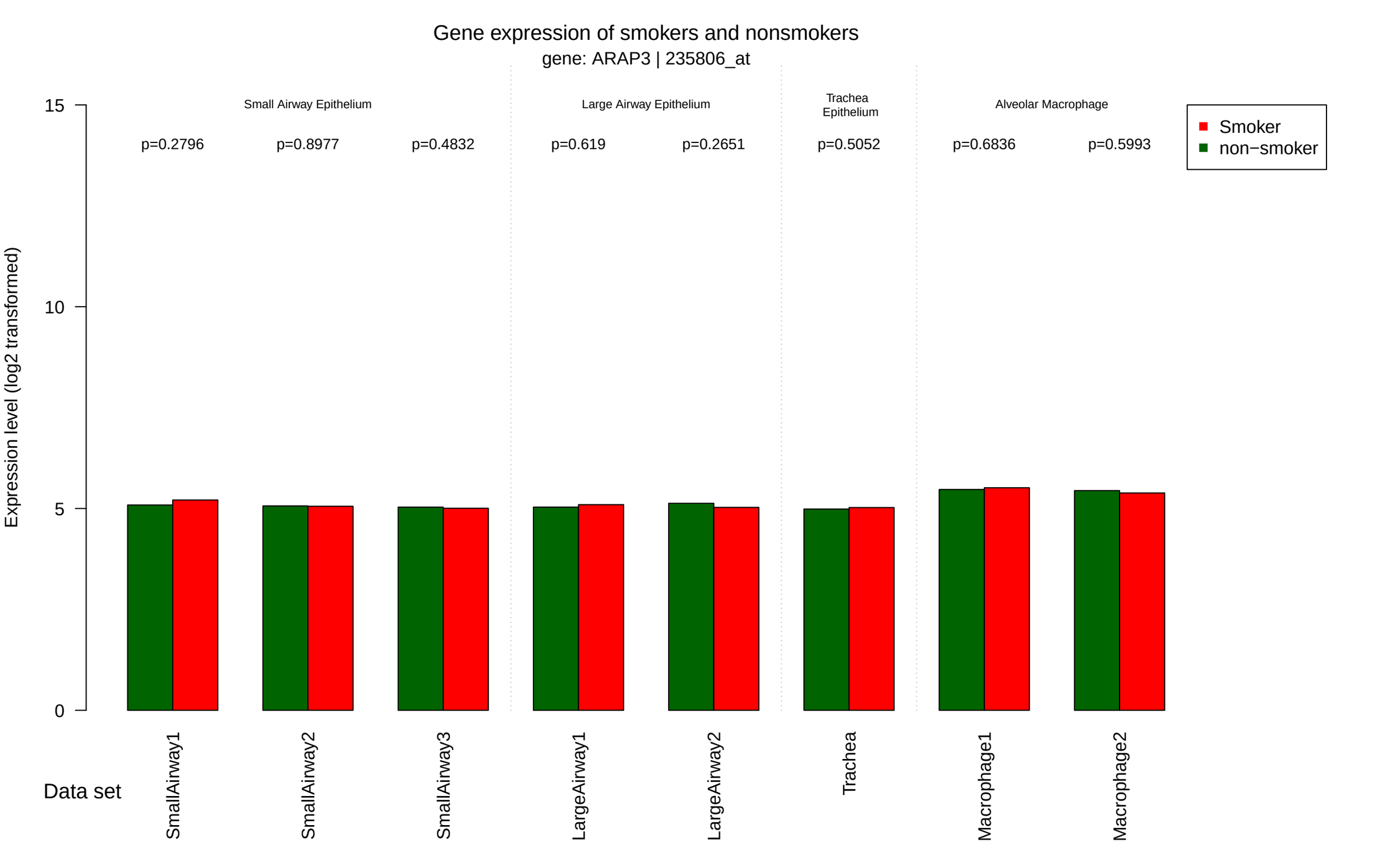
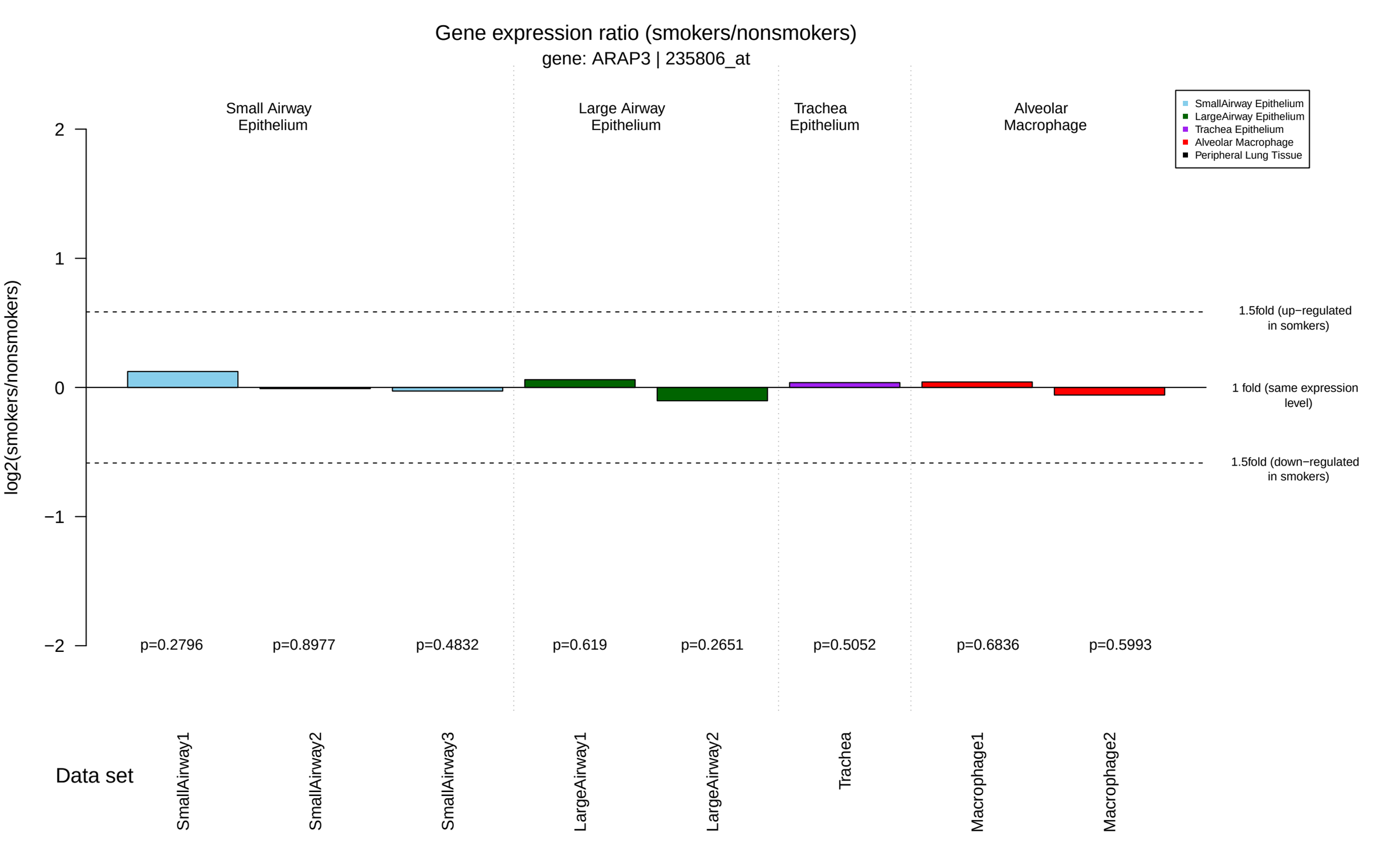
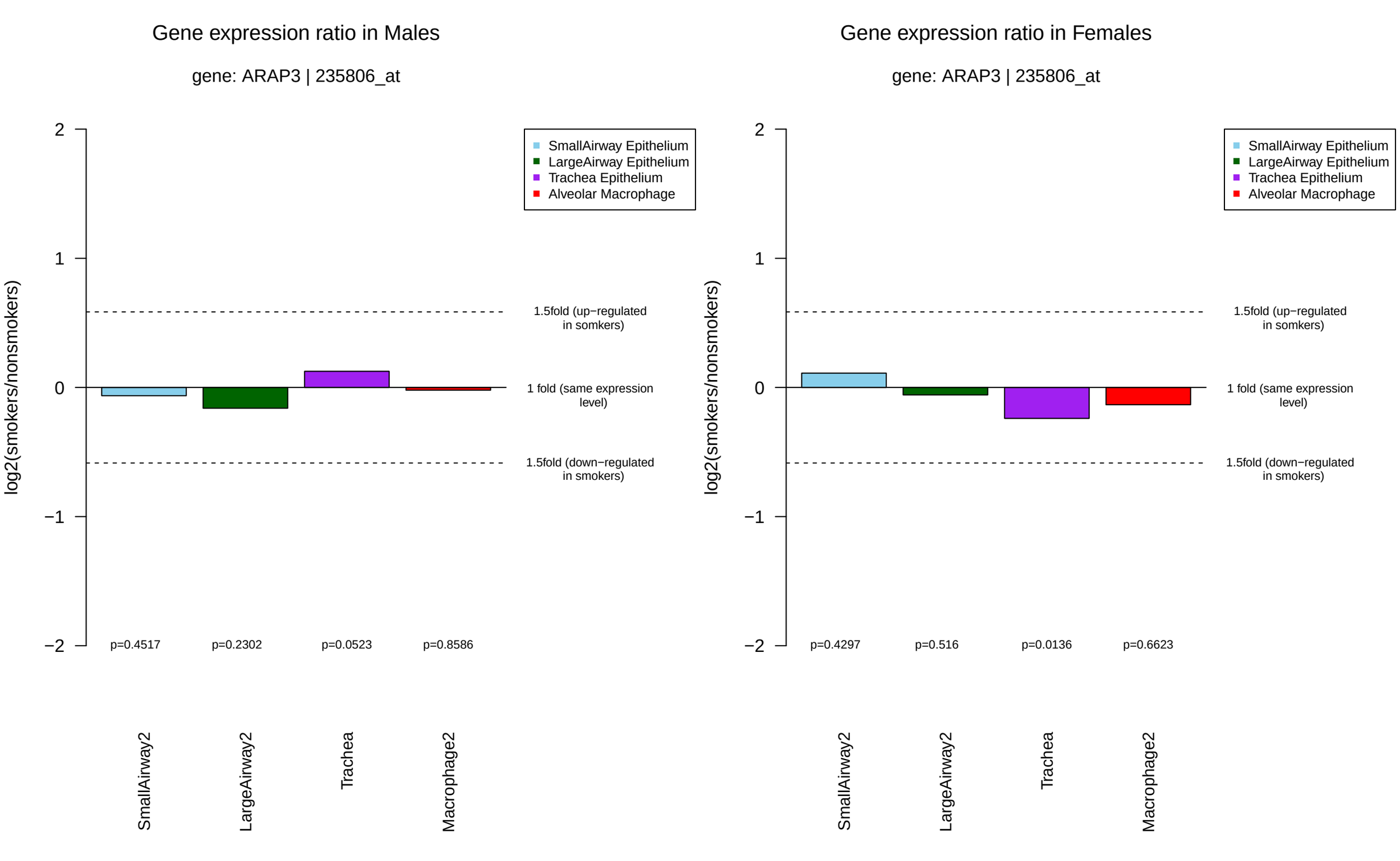
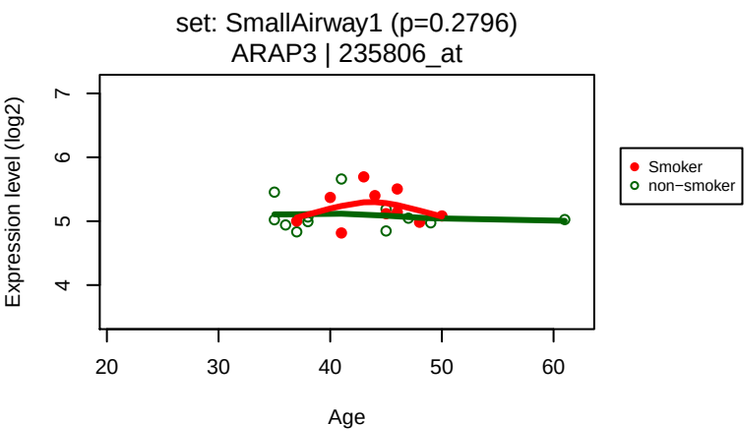
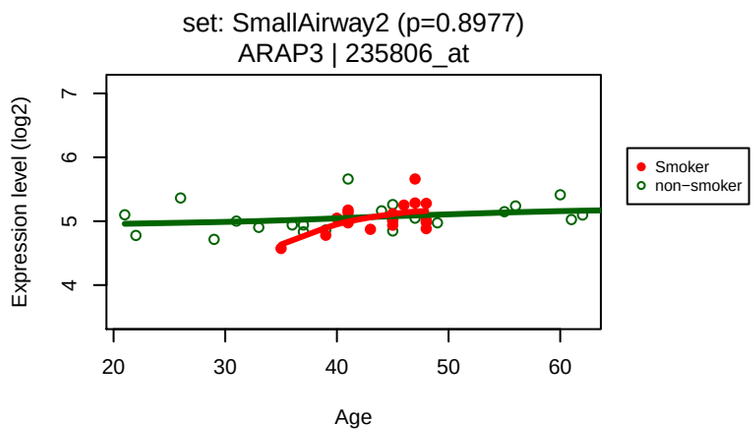
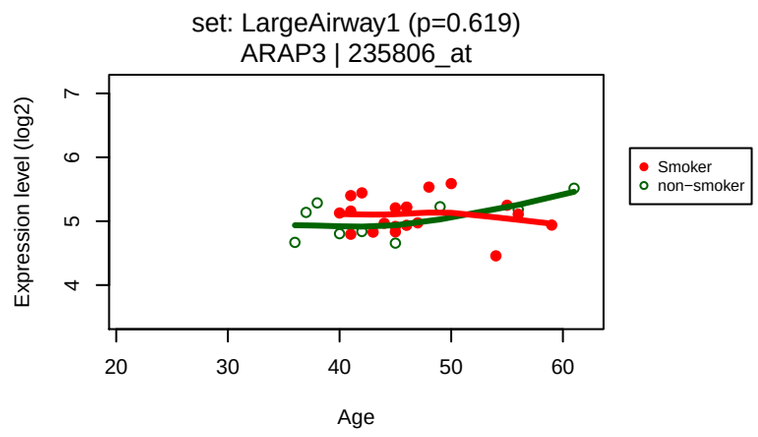

Gene expression of smokers and nonsmokers
gene: ARAP3 | 235806_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2796
p=0.8977
p=0.4832
p=0.619
p=0.2651
p=0.5052
p=0.6836
p=0.5993
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

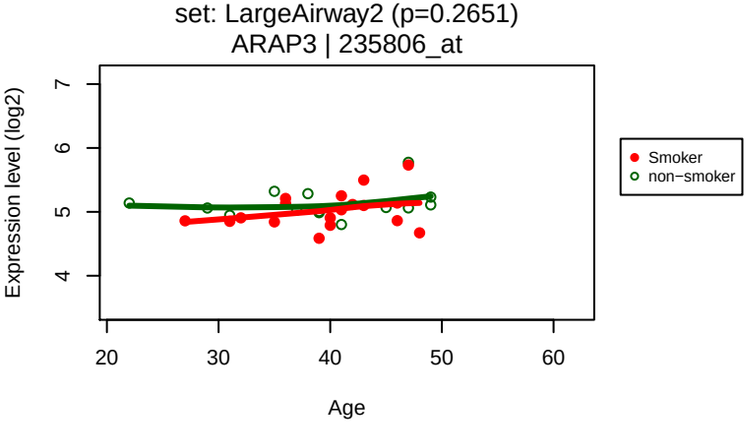

Gene expression ratio (smokers/nonsmokers)
gene: ARAP3 | 235806_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2796
p=0.8977
p=0.4832
p=0.619
p=0.2651
p=0.5052
p=0.6836
p=0.5993
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

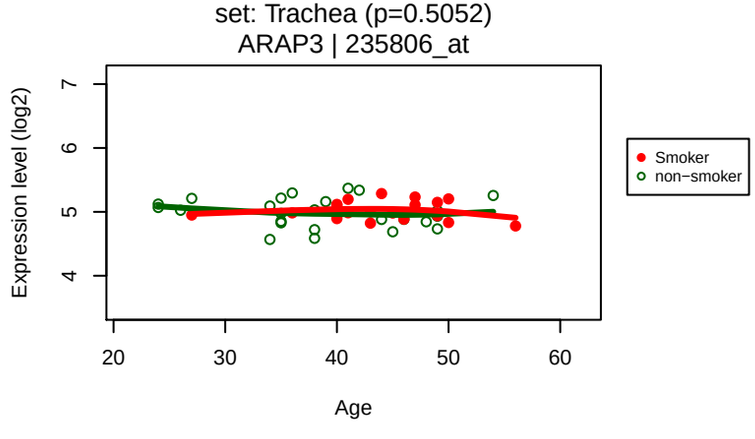

Gene expression ratio in Males
gene: ARAP3 | 235806_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4517
p=0.2302
p=0.0523
p=0.8586
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARAP3 | 235806_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4297
p=0.516
p=0.0136
p=0.6623
SmallAirway2
LargeAirway2
Trachea
Macrophage2

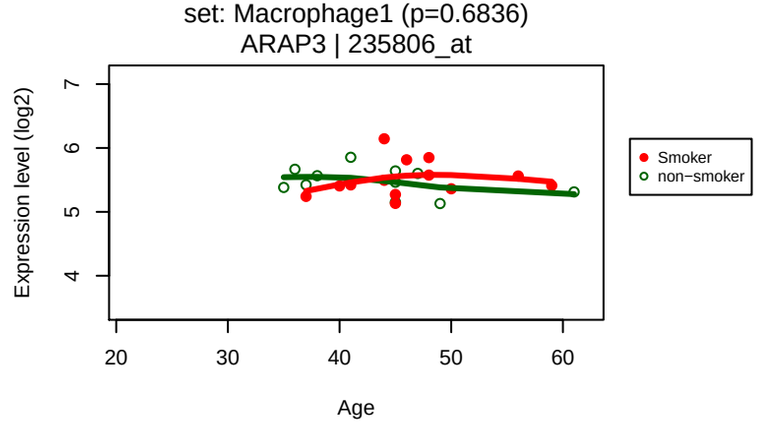

set: SmallAirway1 (p=0.2796)
ARAP3 | 235806_at
Expression level (log2)
Age
Smoker
non−smoker

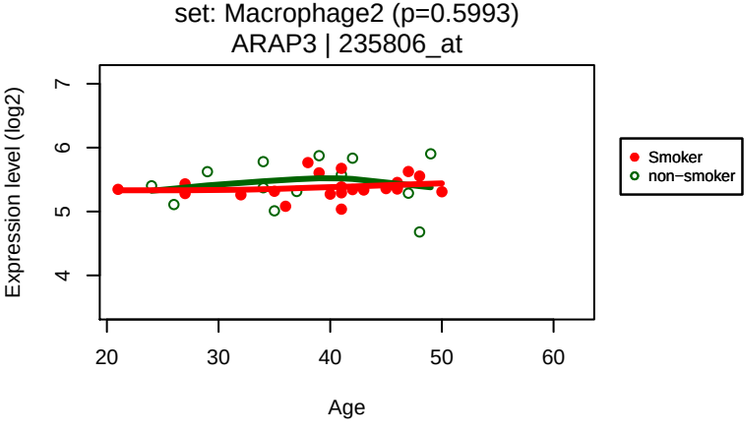

set: SmallAirway2 (p=0.8977)
ARAP3 | 235806_at
Expression level (log2)
Age
Smoker
non−smoker

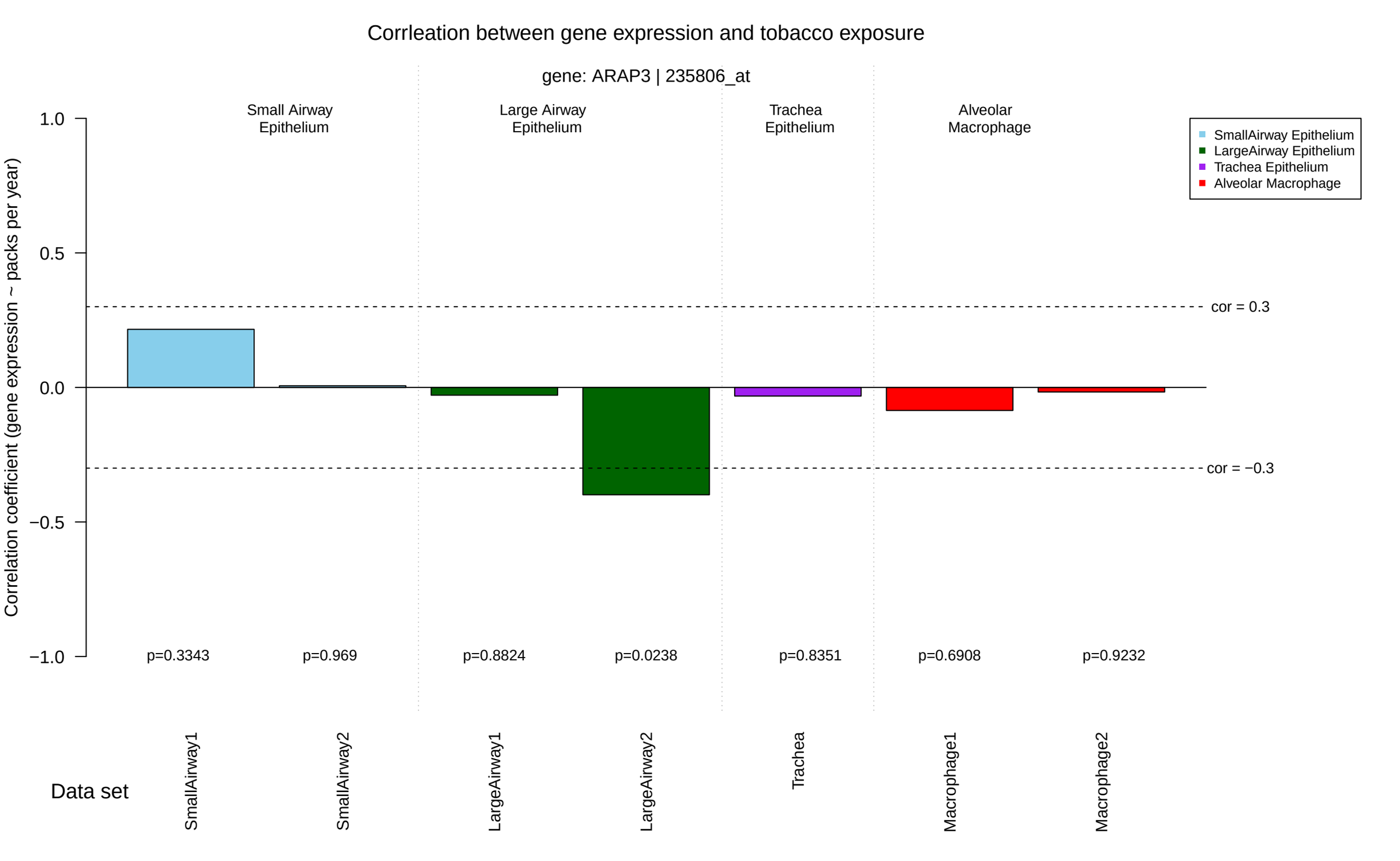

set: LargeAirway1 (p=0.619)
ARAP3 | 235806_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2651)
ARAP3 | 235806_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5052)
ARAP3 | 235806_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6836)
ARAP3 | 235806_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5993)
ARAP3 | 235806_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARAP3 | 235806_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3343
p=0.969
p=0.8824
p=0.0238
p=0.8351
p=0.6908
p=0.9232
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2